

# Cooperation by Default: LLM Agents in the Collective-Risk Dilemma

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Submission Id: «OpenReview submission id»

## Abstract

Language-model agents now act for people in shared settings, where one agent’s choice can put a whole group at risk. We ask whether such agents cooperate by weighing the risk and their partners, or by default: play that does not change with what is at stake. In the collective-risk social dilemma, six players pay into a shared pool over ten rounds, and if the pool misses a target, everyone loses their savings with some probability. For risk-neutral players, the best outcome is to keep the money at low risk and pay a fair share at high risk. We test five low-cost models, one per provider, in almost four thousand games: self-play across risk levels, prompt controls, scripted partners, and tables that mix two models. Four models pay about the fair share at every positive risk level, and four of the five still pay when there is no risk. Removing normative words from the prompt changes neither result. Stating that only the agent’s own cash matters largely stops three models from paying at zero risk, but only one then keeps most of its money at low risk. Asked in a separate call, several models rank the two choices correctly for each risk level, yet their payments do not change. Beside scripted partners who make every payment a loss, no model pays nothing, and two models keep paying after the outcome is settled. In mixed groups, one agent that stops paying in the last round breaks a table of exact fair-share players. Under selection by payoff, scoring agents against their tablemates rewards the smallest payer and lowers group success, while scoring them across separate groups favours the fair-share player at moderate and high risk. Agent evaluations should state the agent’s goal, compare play with the optimum and use mixed populations.

## Keywords

LLM agents; collective-risk social dilemma; cooperation; best response; empirical game-theoretic analysis; multiagent evaluation

### ACM Reference Format:

Anonymous Author(s). 2027. Cooperation by Default: LLM Agents in the Collective-Risk Dilemma. In *Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027)*, Hanoi, Vietnam, 3–7 May 2027, IFAAMAS, 9 pages.

## 1 Introduction

Language-model agents are starting to act for people in shared settings: they spend budgets, share compute, and bargain over common resources. Each such agent is a player in a social dilemma, because what it pays helps the group and costs its owner. Before

deploying one, a designer needs answers to two questions. Does the agent’s cooperation depend on what is at stake? And does it depend on what the other agents do?

Self-play, the most common test, cannot answer either question. When every seat holds a copy of the same model, an agent that reasons about the stakes and an agent that repeats a fixed rule can write the same record of play. Many studies also report whether the group reached its goal, not how far the agents were from the best they could have done. Yet a group of agents can reach its goal and still waste most of its money. Finally, a prompt that gives a group target and a number of rounds implies an equal split, so an agent that repeats that split looks cooperative on every summary measure.

We study these questions in a classic game from research on climate cooperation, the collective-risk social dilemma [16]. Six players pay into a shared pool over ten rounds, and if the pool misses a target, a catastrophe destroys everyone’s savings with probability  $p$ . For risk-neutral players, nobody should pay below a pivot  $p^* = \frac{1}{2}$ , and above it everyone should pay the *fair share*, the target split equally, which is half of each player’s money. At  $p = 0$  paying only lowers a player’s own money, so no attitude to risk can explain it.

We call play a *default* when it does not change with what is at stake. The game gives two sharp tests of a default, because in both paying cannot help: paying at  $p = 0$ , and paying after the outcome can no longer change. We test five low-cost production models, one per provider, in 3,950 games and four designs: self-play across eleven risk levels; prompt controls, including questions about which choice pays more and prompts that remove normative words or state the agent’s goal; one model against five scripted partners, where the best response can be computed exactly; and two different models at one table, which lets us ask which agent would win if agents were selected by payoff [28, 30]. Our findings are four (Figure 1).

- **Risk does not change the default** (Section 4). Four models pay about the fair share at every positive risk level, and four of five still pay at  $p = 0$ . Removing the normative words from the prompt changes neither result. Stating that only the agent’s own cash matters largely stops three models from paying at  $p = 0$ , but only GPT Luna then switches from keeping its money at low risk to paying at high risk. Success rates hide the failures that remain.
- **Answers track the risk; play does not** (Section 5). In separate calls, several models rank the two choices correctly and change that ranking with the risk, but they pay the same amount. Beside scripted partners who make every payment a loss, no model pays nothing, and two models keep paying after the outcome is settled.



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**Figure 1: Overview.** (a) Six players hold 40 units and pay 0, 2 or 4 into a shared pool in each of ten rounds; if the pool misses 120, all savings are lost with probability  $p$ . Below: the best total per seat (Proposition 1). (b) The designs, who sits at each table, and the number of games. (c) The findings; badges mark the design behind each.

- **An exact fair share has no slack** (Section 6). One partner that stops paying in the last round breaks a table of exact fair-share players, while a table that overpays a little absorbs it. Agents pay less when generous partners fill most of the table.
- **The scoring rule decides who is selected** (Section 7). Seats at one table share the pool and the lottery, so the smaller payer always earns more there. Scoring agents against their tablemates therefore favours the smallest payer and, in our games, lowers group success. Scoring them across separate groups favours the exact fair-share player at moderate and high risk.

For the models we test, cooperation is a default that follows from what the prompt leaves unsaid, not a choice that follows from the stakes. A default can work well beside partners like itself, but it cannot be trusted with shared resources, and the way a system states its agents’ goals and selects its agents can make the problem better or worse.

## 2 Related work

*Collective risk and selection.* The collective-risk dilemma was built to study why groups fail to avoid a shared disaster [16]. Human groups reach the target mostly when the risk is high, and inequality and communication change the result [26]. Games with a threshold have many equilibria [18], human contributions decline over the rounds of repeated public goods games [2], and the risk of collective failure can turn a defection trap into a coordination problem [21]. Theory also shows that selection on payoff relative to a small group favours whoever does better than its neighbours [22], while selection between groups can favour cooperation [27]. This literature tells us that behaviour *should* move with the risk, so a flat response is informative, and we reuse its selection theory to read our mixed tables.

*Language models in games.* Language models have been studied as economic subjects [4, 8, 15], in repeated two-player games [1, 9], and in multi-player game benchmarks [11]. Models tend to be more cooperative than people [15], an even split can reflect a focal point rather than fairness [6], and models can hold a correct belief or plan and act against it [7, 23]. In social dilemmas, societies of agents overuse a common resource [20], reasoning models free-ride in public goods games [10], and agents replace people in a collective-risk

game [24]. Closest to us, Willis et al. study the same collective-risk game with strategies written by three models, map welfare over mixtures of selfish and collective strategies, and find that payoff-biased imitation drives larger groups to selfish, low-welfare outcomes [31]. We differ in three ways. We record the turn-by-turn play of deployed models, not code they write. We price that play against a computed optimum, with two tests where paying cannot help. And we show that whether selection compares tablemates or separate groups decides which model wins, which gives one simple route to selfish outcomes.

*Multiagent evaluation.* Sequential social dilemmas made cooperation a property of learned policies [12, 14], and cooperative AI asks for agents that work with partners unlike themselves [5, 25]. Melting Pot tests a focal population against background agents it has never met [13]. Empirical game-theoretic analysis treats trained agents as strategies of a meta-game [28, 30], and  $\alpha$ -Rank ranks them by evolutionary dynamics [17]. Tournaments that mix language models with classic strategies report model-specific strategic fingerprints in the prisoner’s dilemma [19], and cooperation evolves differently across base models in a donor game [29]. We add a many-player dilemma in which the best response to fixed partners can be computed exactly, and real tables of two different models.

*Overlap with concurrent work.* A concurrent manuscript uses the same game and an overlapping set of models to ask whether models reproduce the risk sensitivity of human players, and suggests that more capable models can respond to risk [3]. It reports a self-play risk baseline related to part of Section 4. The present paper asks how low-cost agents behave against the optimum, against fixed partners and against other models; the tests where paying cannot help and Sections 5 to 7 have no counterpart there.

## 3 The game and what rational play looks like

### 3.1 Rules

Six players each start with  $e = 40$  units. A *seat* is one player position and a *table* is the six seats of one game. In each of  $R = 10$  rounds, every player privately chooses a contribution  $a_i^t \in \{0, 2, 4\}$  to a shared *pool*; after each round all players see every seat’s past contributions. Let  $x_i = \sum_t a_i^t$  be a player’s total and  $X = \sum_i x_i$  the pool. The target is  $T = 120$ . A player keeps what it did not pay, but if the pool misses the target a catastrophe destroys all savings with

probability  $p$ :

$$u_i = \begin{cases} e - x_i & \text{if } X \geq T, \\ 0 \text{ with probability } p, \text{ else } e - x_i & \text{if } X < T. \end{cases} \quad (1)$$

The fair share is  $T/6 = 20$  per player, or 2 per round.

**Proposition 1.** *Consider the game in which each player picks a total  $x_i \in \{0, 2, \dots, 40\}$ , and let  $p < 1$ . A profile is a Nash equilibrium if and only if  $x = 0$ , or  $X = T$  and  $x_i \leq pe$  for every  $i$ . A target-reaching equilibrium exists if and only if  $p \geq p^* = T/(6e) = \frac{1}{2}$ . The welfare-optimal outcome is  $x = 0$  for  $p < p^*$  and  $X = T$  for  $p > p^*$ , so the best attainable payoff per player is  $\text{opt}(p) = \max\{(1-p)e, e - T/6\}$ .*

*Proof sketch.* If  $X < T$ , any player with  $x_i > 0$  gains by paying nothing, so the only equilibrium that misses the target is  $x = 0$ . If  $X > T$ , a player can pay 2 less and still reach the target. If  $X = T$ , the best deviation is to pay nothing, worth  $(1-p)e$  against  $e - x_i$ , which gives  $x_i \leq pe$ ; summing over players gives  $p \geq T/(6e)$ . Welfare is  $6e - X$  when the target is met and  $(1-p)(6e - X)$  otherwise, so it peaks at  $X = T$  or  $X = 0$ .  $\square$

The ten-round game, whose strategies can react to the history, has the same totals in pure-strategy Nash equilibrium; we claim nothing about subgame perfection. Paying nothing in every round secures  $(1-p)e$  whatever the others do, so no equilibrium misses the target with  $x \neq 0$  or has  $x_i > pe$ . A payment beyond the target can be dropped in the round the target is reached, or later, and nothing played afterwards can lower that player’s payoff. Each profile of Proposition 1 is played by fixed schedules that ignore the history.

Three benchmarks follow and carry the rest of the paper. First, the *optimum*, the welfare optimum for risk-neutral players, switches at  $p^*$ , as does the best response to five fair-share partners; it is a benchmark, not a prediction, since  $x = 0$  is an equilibrium at every  $p$ . Second, paying is *dominated*, meaning that every unit paid lowers the seat’s own payoff whatever else happens, in two situations: at  $p = 0$ , where both branches of (1) are equal, and once the outcome is *settled*, meaning that the visible history already proves that more payment cannot change it (the pool has reached the target, or cannot reach it even if every seat pays 4 in every remaining round). Neither test depends on the agent’s attitude to risk or on its beliefs. Third, these two tests are the sharpest evidence of a default: payment that continues where paying is dominated cannot follow from the stakes.

**Proposition 2.** *For two seats  $i$  and  $j$  in the same game, the difference in their expected payoffs over the lottery is  $c(x_j - x_i)$ , with  $c = 1$  if the target was reached and  $c = 1 - p$  otherwise.*

Both seats face the same pool and the same lottery, so only their own payments differ. Hence, for  $p < 1$ , whoever pays less at a table earns more, whatever the group achieves.

### 3.2 Models, designs and measures

We test five production models, one per provider, all at the low-cost tier of their family: Claude Haiku 4.5 (Anthropic), Gemini 3.5 Flash-Lite (Google), GPT-5.6 Luna (OpenAI), Qwen3-235B-A22B-Instruct (Alibaba) and Grok 4.20 in its non-reasoning mode (xAI). We call them Claude Haiku, Gemini Flash-Lite, GPT Luna, Qwen and Grok; the supplementary material gives the exact model identifiers. We

**Table 1: Designs.** A cell is one condition at one risk level: one model in self-play, one model with one kind of script, or one pair of models with one split of seats. Each cell holds ten games. The in-game question condition uses the unchanged prompt, so it also repeats the self-play grid at three risk levels.

| Design                              | Risk levels $p$    | Games |
|-------------------------------------|--------------------|-------|
| Self-play grid                      | 0, 0.1, ..., 1     | 550   |
| In-game questions                   | 0.1, 0.5, 0.9      | 150   |
| No hint                             | 0.1, 0.5, 0.9      | 150   |
| Neutral wording                     | 0, 0.1, 0.9        | 150   |
| Neutral wording, goal stated        | 0, 0.1, 0.9        | 150   |
| Two rewordings                      | 0.1, 0.9           | 200   |
| Temperature zero                    | 0.1, 0.9           | 100   |
| One model, five scripts (4 kinds)   | 0.1, 0.3, ..., 0.9 | 1,000 |
| Two models (10 pairs, 1 to 5 seats) | 0.1, 0.5, 0.9      | 1,500 |
| All                                 |                    | 3,950 |

chose this tier because agents that act at scale are likely to run on it. Table 1 lists the designs. Every model plays the same number of games in every design, and every cell holds 10 games.

*Prompts.* Each round a seat receives one prompt: the rules, the full history of every seat’s contributions, and a request for one contribution. The supplementary material gives every prompt verbatim. The base prompt calls the pool a “climate account”, says that the group “must” reach the target, calls the loss a “disaster”, names the game as a collective-risk social dilemma, and never states the player’s goal. It also states the equal split twice, as an average of 2 per round and as an example in which paying 2 every round leaves 20. The prompt controls change exactly these parts. *No hint* deletes the two equal-split phrases. *Neutral wording* replaces the four normative or naming phrases with plain ones, such as “a shared account” and “the outcome depends on whether” the pool reaches 120. *Goal stated* keeps the neutral wording and adds an opening that says the player is in a paid experiment where “the only thing that matters to you is your own final cash payoff”. Two *rewordings* change the phrasing but keep every rule and the hint.

*Protocol.* The running pool is not printed, so a seat that wants it must add up the history. The prompt is in English, no seat is given a persona, and the temperature is 0.7 unless stated. A reply is read from its final CONTRIBUTION: line; replies cut off at the output limit are retried with a larger limit rather than guessed. The data contain 187,000 model decisions. Only 2 replies could not be read, and both come from one reworded prompt.

*Measures.* We report the *expected* payoff, the average a seat would earn if the catastrophe lottery were run many times on the same contributions, and the share of  $\text{opt}(p)$  it loses. We do not use realised payoffs. A catastrophe strikes when a uniform random draw falls below  $p$ , and the draw depends only on the repetition number, so our games reuse ten draws in every cell, all below 0.47. Every missed game at  $p \geq p^*$  therefore ended in catastrophe, and realised payoffs would punish models that miss more than chance would. Agents never see the draw, so their play is not affected. A

seat plays the *exact split* if it pays 2 in all ten rounds. The unit of analysis is the game, since seats in one game share a history and a lottery. Intervals are 95% percentile bootstraps over games and tests are permutation tests over games.

#### 4 Risk does not change the default

Four of the five models pay about the fair share at every positive risk level (Figure 2). For Claude Haiku, Gemini Flash-Lite, GPT Luna and Qwen, the mean total below the pivot and the mean total above it differ by at most 0.2 units, while a risk-neutral player would jump from 0 to 20. Risk aversion does not explain the flat line either: preferring a certain 20 to keeping 40 with probability 0.9 requires a coefficient of relative risk aversion of at least 0.84. Gemini Flash-Lite is the clearest case: 99.7% of its seats play the exact split at every risk level, so its play is optimal above the pivot only because a fixed rule matches the optimum there.

*Paying when there is nothing to protect.* Four of the five models pay at  $p = 0$ , close to their usual amount, although paying there only lowers a player’s own money (240 of 300 seats pay). GPT Luna is the one exception: it pays nothing at  $p = 0$  and its usual share at every positive risk, so it separates no risk from some risk but does not grade the risk.

*Three ways to lose.* The models lose payoff in three ways (Table 2). Gemini Flash-Lite, Claude Haiku and GPT Luna lose mostly below the pivot, where they pay for a target that is not worth reaching. Grok pays far more than needed: its groups reach the target by round 6.7 on average, and each seat then pays another 13.0 units into a pool that is already safe. Qwen fails at the end. It pays 2 in 99% of decisions in rounds 1 to 9 and then 95% of its seats pay nothing in round 10, so its groups reach the target in only 2 of 110 games; had each seat repeated its round-9 move, 100% would have succeeded. This is a coordination failure, not one bad move: with 108 in the pool before round 10, one seat alone can add at most 4, so paying nothing is also an equilibrium of that round, and Qwen seats almost always pick it, much as human contributions decline at the end of repeated public goods games [2]. GPT Luna shows a milder form: its groups miss the target in 23 of 100 games with  $p > 0$ , always after keeping pace until the last round. Claude Haiku shows that the pool can be read from the same prompt: it pays nothing in round 10 in 89% of seats once the pool has reached the target, against 12% when it has not. Success rates hide all of this. Gemini Flash-Lite and Grok both reach the target in every game, yet Grok keeps a quarter as much money, and the panel loses 42% of the attainable payoff.

*The default survives rewording, but its failures do not.* Deleting the two equal-split phrases does not lower anyone’s payment (Table 3), so the hint was a ceiling and a guide for precision, not the source of the cooperation. Gemini Flash-Lite still plays the exact split, which it can rebuild from the target, and GPT Luna keeps its average but not its precision (the exact split falls from 83% to 56% of seats). Qwen and Grok pay more; Qwen now reaches the target in 100% of games instead of 0%, but earns less because it overpays. Two rewordings and a temperature of zero leave the mean payments of the other four models within 1.0 unit of the grid. The failures are less stable than the means: under the second rewording Grok

**Table 2: Self-play summary over the risk grid, using expected payoffs. Pays at  $p=0$  is the mean seat total where paying is dominated. Exact split is the share of seats paying 2 in all ten rounds. Target is the share of games that reach it, and Payoff is the expected payoff per seat; the optimum averages 25.5 over the grid. Lost is the share of the attainable payoff  $\text{opt}(p)$  that was lost, and Main loss is its largest cause: paying toward the target below  $p^*$ , paying beyond 120, or missing the target at or above  $p^*$ , which for Qwen happens in the last round.**

| Model             | Pays at $p=0$ | Exact split (%) | Target (%) | Payoff | Lost Main (%) loss |
|-------------------|---------------|-----------------|------------|--------|--------------------|
| Claude Haiku      | 20.9          | 17              | 98         | 18.9   | 26 below $p^*$     |
| Gemini Flash-Lite | 20.0          | 99.7            | 100        | 20.0   | 21 below $p^*$     |
| GPT Luna          | 0.0           | 72              | 70         | 19.2   | 25 below $p^*$     |
| Qwen              | 18.0          | 4               | 2          | 10.9   | 57 last round      |
| Grok              | 37.4          | 2               | 100        | 5.0    | 80 overshoot       |

pays the fair share, and the share of Qwen seats that pay nothing in round 10 ranges from 41% to 100% across prompts. Grok’s rise with risk on the grid (6.2 units) shrinks to 1.9 in the rerun, so we do not read it as a response to risk.

*Stating the goal, not removing the norms, switches the default off.* The default does not come from the normative words. With the neutral wording, four models still pay at  $p = 0$  (245 of 300 seats pay), and at  $p = 0.1$  and  $p = 0.9$  four models pay what they paid before while Grok pays even more (Table 3). Adding the stated goal changes play at zero risk: Claude Haiku, Gemini Flash-Lite and Qwen then pay 0.3, 1.4 and 4.8 units at  $p = 0$ , against 20.9, 20.0 and 18.0 with the base prompt. The goal does not make most models weigh the risk. Claude Haiku and Gemini Flash-Lite still pay the fair share at  $p = 0.1$  (19.8 and 20.2 units), so they separate no risk from some risk but do not grade it. GPT Luna alone grades it, paying 7.7 units at  $p = 0.1$  and 20.1 at  $p = 0.9$ . Qwen pays less at every level and reaches the target in 0 of 30 games, and Grok keeps paying far more than needed. The default is therefore what these models do when the prompt leaves their goal unstated; the scripted and mixed designs below use the base prompt and describe it.

#### 5 Answers track the risk; play does not

A flat response to risk has a cheap explanation: perhaps the agents do not understand the game. We test this in two ways. We ask the playing agent which choice pays more, and we seat it with partners whose behaviour is fixed, so that the *best response*, the play that earns the most against them, can be computed exactly.

##### 5.1 Asking the agent

In 150 games we question the first seat in rounds 1, 5 and 10, on the state it has just decided on. Each question is a separate call, and its answer never enters the game. Of the 4,500 questions, 3,600 ask about the rules, such as the target or the allowed amounts. Agents answer 100% of them correctly, but these questions only ask for numbers printed in the prompt, so they show that agents read the

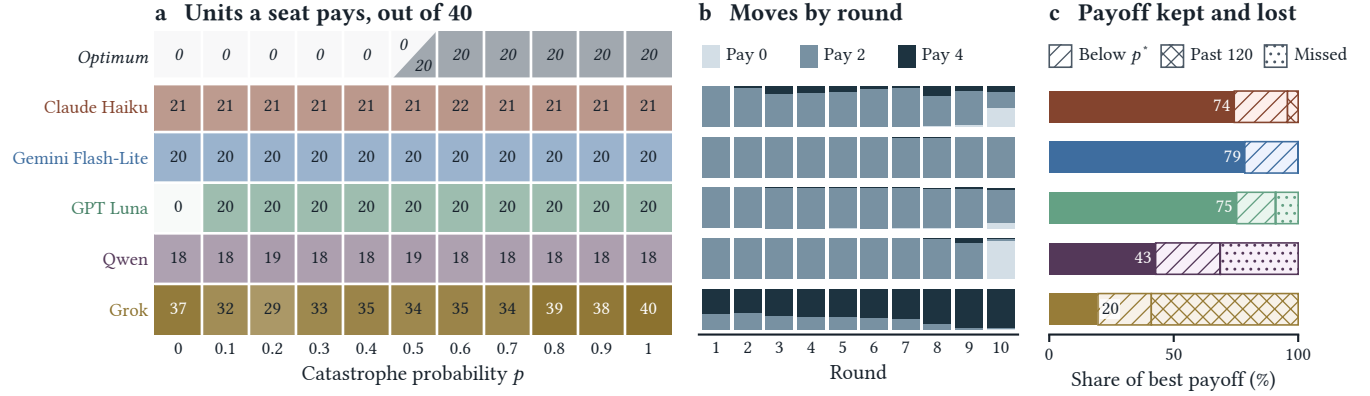


Figure 2: Self-play on the risk grid, 110 games per model. (a) Mean units a seat pays out of 40, ten games per cell; the top row is the optimum of Proposition 1, where both 0 and 20 are optimal at  $p = 0.5$ . (b) Share of seats paying 0, 2 or 4 in each round, over the 100 games with  $p > 0$ . (c) Expected payoff kept as a share of  $\text{opt}(p)$ , and the loss split into paying toward the target below  $p^*$ , paying past 120, and missing the target at or above  $p^*$ .

Table 3: Mean seat total under prompt and decoding changes, pooled over  $p = 0.1$  and  $0.9$  (20 games per entry). *Base* is the self-play grid. *Rerun* is the in-game question condition, which uses the unchanged prompt and was collected with the other changes. *No hint* deletes the equal-split phrases, *Para. 1* and *Para. 2* are the two rewordings, *Temp. 0* sets the temperature to zero, *Wording* uses the neutral wording, and *Neutral* adds the stated goal to it. Bold marks a shift from *Base* of at least 2 units with permutation  $P < 0.01$ . Temperature zero was not deterministic: a cell produced 5.5 distinct games in ten, against 6.6 at 0.7.

| Condition | Claude Haiku | Gemini Flash-Lite | GPT Luna    | Qwen        | Grok        |
|-----------|--------------|-------------------|-------------|-------------|-------------|
| Base      | 20.9         | 20.0              | 19.9        | 18.4        | 35.1        |
| Rerun     | 21.3         | 20.0              | 19.9        | 18.5        | 35.6        |
| No hint   | 21.7         | 20.0              | 19.9        | <b>34.7</b> | <b>40.0</b> |
| Para. 1   | 21.3         | 20.0              | 19.9        | 19.4        | 31.8        |
| Para. 2   | 20.9         | 20.0              | 19.8        | 18.0        | <b>20.6</b> |
| Temp. 0   | 21.2         | 20.0              | 20.0        | 18.3        | 35.1        |
| Wording   | 21.0         | 20.0              | 19.4        | 18.7        | <b>39.7</b> |
| Neutral   | 20.0         | 20.1              | <b>13.9</b> | <b>8.2</b>  | 37.7        |

rules, not that they understand them. We therefore rest on one value question, asked verbatim:

*Compare two strategies for the whole game. (A) You contribute 2 in every one of the 10 rounds and the group reaches the target, so you finish with 20 for certain. (B) You contribute 0 in every round and the group does not reach the target, so you keep 40 unless the disaster strikes. Which gives the higher AVERAGE final cash? Answer 1 if A is higher, 2 if B is higher, 0 if they are exactly equal.*

The correct answer is B below  $p^*$ , A above it, and equal at  $p^*$ . A second value question could be read in two ways and is not used.

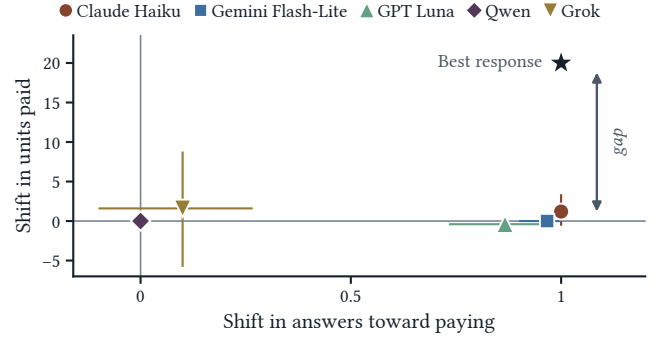
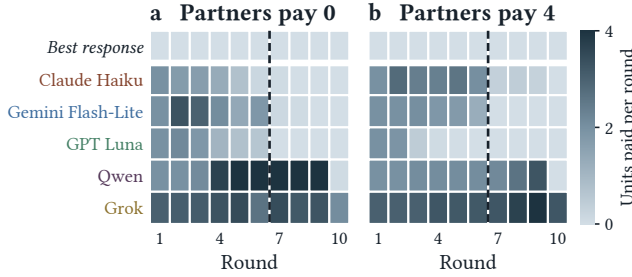


Figure 3: What agents say and do, from  $p = 0.1$  to  $p = 0.9$  in the same games (ten at each), with 95% intervals. Horizontal: drop in the share of answers saying that paying nothing pays more on average; a correct answerer moves by 1. Vertical: change in what the questioned seat pays. Star: a best responder beside fair-share partners.

*Answers move with the risk; play does not.* Claude Haiku and Gemini Flash-Lite answer the value question correctly almost every time, and GPT Luna in 84% of answers. All three change their answer with  $p$  (Figure 3, horizontal axis): in rounds 1 and 5 at  $p = 0.1$  they say that keeping pays more in 97% of answers, and at  $p = 0.9$  they mostly say the opposite. Their payments do not follow (Figure 3, vertical axis): the same seats pay 20.1 units at  $p = 0.1$  and 20.3 at  $p = 0.9$ . Qwen and Grok do not show the knowledge at all, with 33% and 46% correct. Qwen says that keeping pays more in 100% of its answers at every risk level, and Grok gives that answer in round 1 whatever the risk. The question lays out both strategies, which is easier than finding the comparison inside the game, so the result is a gap between answers and play, not evidence about what an agent computes when it decides.



**Figure 4: One model beside five scripted partners, pooled over five risk levels (50 games per row). A cell is the mean payment in one round; the top row is the best response, 0 in every round. The dashed rule follows round 6, after which the outcome is settled in most games.**

## 5.2 Playing against fixed partners

Next we seat one model with five scripted partners of one kind: they always pay 0, always pay 2, always pay 4, or copy the group, starting at 2 and then paying the rounded average of the other seats’ previous round. The model is not told that its partners are scripted, but their moves are in the history. We found the best response to each kind by trying all  $3^{10}$  sequences of the model’s own moves. Beside partners who always pay 0 the target cannot be reached, and beside partners who always pay 4 it is reached without the model, so the best response to both is to pay nothing. Paying becomes dominated, in the sense of Section 3, only once the history settles the outcome. Beside fair-share partners the best response is to pay nothing below  $p^*$  and exactly 20 above it; the copying partners never leave 2 and give the same answer.

*No model pays nothing where that is the best response.* In none of the 500 games beside these two kinds of partner does the model pay nothing; the smallest total we see is 2 units. Only 186 of 1,000 games are exactly optimal, and every one of them has a total of 20, the fair share. The models land on the optimum only where the optimum is the default.

*Partners matter; risk does not.* We split the variation in what a model pays beside the three fixed-amount partners into parts due to the partner, the risk and their interaction. For Claude Haiku, Gemini Flash-Lite, GPT Luna and Qwen, the kind of partner explains 69% to 92% of the variation, and the risk at most 5%. For an exact best responder in the same games, the risk would explain 50%. Beside fair-share partners, where the best response jumps from 0 to 20 at  $p^*$ , no model’s total differs by more than 2 units between low and high risk. Grok’s total follows neither the partner (3%) nor the risk (8%). Its game is decided by its first move, which it samples before any history exists: it opens with 4 in 103 of 200 games and mostly repeats that opening, and this first move alone explains 74% of the variation in its total.

*Responses to partners take simple forms.* The models respond to their partners, but not the way a best responder would (Figure 4, Table 4). Gemini Flash-Lite and Claude Haiku pay their usual rate and then stop once the history shows that the outcome is settled. GPT Luna pays little even before that point, above all beside partners

**Table 4: Units a model pays per game before and after the outcome is settled, beside five partners who always pay 0 or always pay 4 (50 games per entry). Given these partners, every unit paid is wasted; units paid after settlement are wasted even for an agent that does not know its partners are scripted.**

| Model             | Always 0 |       | Always 4 |       |
|-------------------|----------|-------|----------|-------|
|                   | Before   | After | Before   | After |
| Claude Haiku      | 7.7      | 0.1   | 14.2     | 1.1   |
| Gemini Flash-Lite | 13.6     | 0.2   | 10.9     | 0.0   |
| GPT Luna          | 8.5      | 0.0   | 4.5      | 0.0   |
| Qwen              | 17.8     | 12.1  | 12.3     | 8.1   |
| Grok              | 19.6     | 11.1  | 16.7     | 16.2  |

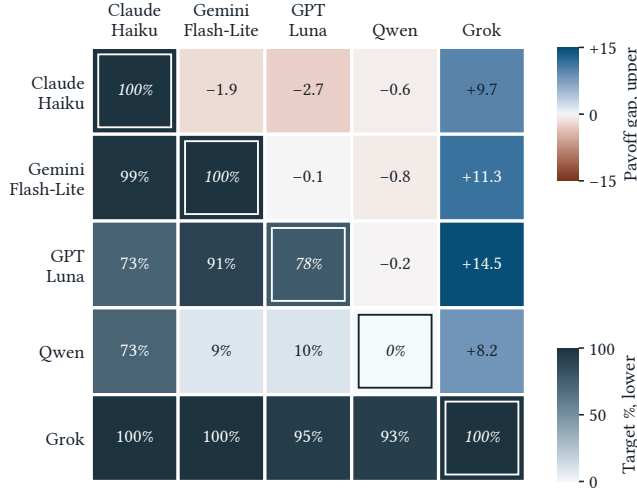
who always pay 4, which is close to a best response while it learns who its partners are. Qwen raises its payment to 4 beside partners who pay nothing. Qwen and Grok keep paying large amounts after the outcome is settled, which no belief about the partners can justify. Qwen’s last-round rule also costs its partners: beside five fair-share partners it pays 2 for nine rounds and then 0, and the table misses the target in 47 of 50 games, each time by exactly 2 units. Gemini Flash-Lite in the same seat reaches the target in 50 of 50.

## 6 Defaults in mixed groups

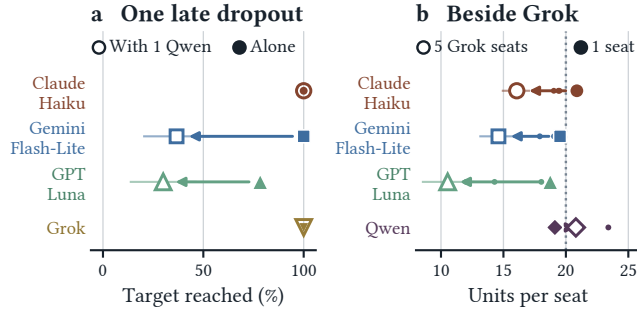
Self-play and scripted partners cannot show what happens when agents with different defaults share a table. We seat every pair of models together, with one to five seats for the first model, at  $p \in \{0.1, 0.5, 0.9\}$  and ten games per cell, for 1,500 games. The seats of each model form one block whose position is fixed within a cell and alternates across cells, so every model holds every seat equally often overall.

At a shared table every model earns more than Grok, by up to 14.5 units per seat, while the payoff gaps among the other four are small (Figure 5, above the diagonal). This follows from Proposition 2: Grok pays the most, and at one table the larger payer earns less. Group success depends on the pair (Figure 5, below the diagonal). Tables that include Grok almost always reach the target, while tables that mix Qwen with Gemini Flash-Lite or GPT Luna rarely do; Gemini Flash-Lite and Qwen succeed in only 9% of their games.

*An exact fair share has no slack.* One partner that stops paying in the last round is enough to break a table of exact fair-share players (Figure 6a). When the pool stands at 108 before round 10, Qwen pays nothing in 92% of its decisions, against 0.1% for Gemini Flash-Lite. Five Gemini Flash-Lite seats with one Qwen seat therefore succeed in only 11 of 30 games, although Gemini Flash-Lite alone never misses, and five GPT Luna seats with one Qwen seat succeed in 9 of 30, against 78% alone. Claude Haiku and Grok tables, which pay more than needed, absorb the same partner (30 of 30 for Claude Haiku). In 92% of the 323 missed tables with a Qwen seat, turning its last-round zeros into 2s would have reached the target. Slack also works the other way: one Grok seat lets five Qwen seats reach the target in 25 of 30 games, and Grok pays for the rescue. Since Qwen’s last-round rule depends on the wording, this measures



**Figure 5: Two models at one table, 150 games per pair, pooled over seat splits and risk. Above the diagonal: expected payoff of the row model’s seats minus the column model’s, in the same games. On and below it: share of games reaching the target; boxed cells are self-play (60 games).**



**Figure 6: Two models at one table, pooled over  $p$ . Arrows run from the filled marker to the open one, which carries a 95% interval. (a) Target reached by six seats of one model (60 games) and by five seats plus one Qwen seat (30). (b) What a seat pays beside one and five Grok seats (30 games each; dots: two to four); dotted: the fair share.**

what one late dropout does to a group, whichever model plays that role.

*Generous partners are under-matched.* Agents pay less, not more, when generous partners fill most of the table (Figure 6b). Next to five Grok seats, GPT Luna pays 10.5 units per seat, against 19.1 next to other partners, and Gemini Flash-Lite and Claude Haiku pay 14.6 and 16.1. As with scripted partners, Gemini Flash-Lite and Claude Haiku stop once the pool reaches the target, while GPT Luna pays less from the start, so the generous partner carries the group. Qwen pays slightly more beside Grok. These responses fit watching the pool, which our design does not separate from tracking who paid.

## 7 Which agent wins under selection

Systems that host many agents keep some and drop others: an operator picks the agent that earned most, or users copy the one that did well. The key question is whom an agent is compared with, and we show that the answer decides which of our five models such a process favours. Following empirical game-theoretic analysis [28, 30], each model is a strategy. We write  $\pi_A(k)$  for the mean expected payoff of an  $A$  seat at a table with  $k$  seats of  $A$  and  $6 - k$  seats of another model  $B$ , estimated from the mixed tables for  $1 \leq k \leq 5$  and from self-play for  $k = 6$ .

*Selection process.* A population of  $N$  agents holds  $i$  agents of type  $A$  and  $N - i$  of type  $B$ , and tables of six are drawn at random. The fitness of each type is its payoff averaged over the tables it can land in, as given by

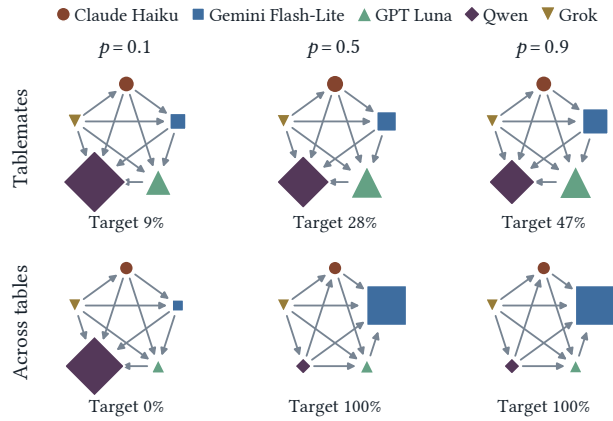
$$f_A(i) = \sum_{k=1}^6 \mathcal{H}(k-1; N-1, i-1) \pi_A(k), \quad (2)$$

$$f_B(i) = \sum_{k=0}^5 \mathcal{H}(k; N-1, i) \pi_B(6-k),$$

where  $\mathcal{H}(k; N-1, m)$  is the hypergeometric chance that  $k$  of the five other seats come from the  $m$  other agents of type  $A$ . An agent copies a random other agent with probability  $1/(1 + e^{-\beta \Delta f})$ , where  $\Delta f$  is their fitness gap and  $\beta$  is the selection intensity, set to 1 per payoff unit.  $\alpha$ -Rank [17] assumes that a new model enters rarely, so the population is almost always made of one model. It computes the chance that a single newcomer takes over, for every pair of models, and turns these chances into a Markov chain over the five one-model populations. The share of time the chain spends on each model is the weight selection gives it, its *stationary mass*. The population size sets whom an agent is compared with. With  $N = 6$  the whole population is one table, so an agent is compared only with its tablemates. With  $N = 30$ , fitness is averaged over random tables, so an agent is compared mostly with agents who sat elsewhere.

*Scoring against tablemates rewards the smallest payer.* Within one table, Proposition 2 fixes the ranking before any data are seen: the seat that pays less earns more. What the data add is who pays least in company, which is Qwen, and what this selection costs. It gives Qwen the largest weight at every risk level (Figure 7, top row). The selected population then reaches the target in only 28% of games at  $p = 0.5$  and 47% at  $p = 0.9$ , against 75% and 76% for a model picked at random. With stronger selection ( $\beta = 10$ ) the success rate falls to 0% and 1%. This is the known result that selection on payoff relative to a small group favours whoever does better than its neighbours, even at a cost to the group [22], now shown with language-model agents.

*Scoring across tables keeps the fair-share player.* When agents are compared with agents at other tables, what matters is how well a model’s own tables do. At  $p = 0.5$  and 0.9 selection then favours Gemini Flash-Lite (Figure 7, bottom row), whose population reaches the target in 100% of games at  $p = 0.9$  and earns the optimum, 20.0 units per seat. It wins because it is rigid: a newcomer that pays less pushes its table below the target, and one that pays more loses money. At  $p = 0.1$  both rules favour Qwen, which still earns far less than the optimum, since every model pays at low risk. The winners do not change with stronger selection, with populations of 60 and 120 agents, or when self-play payoffs use only the risk



**Figure 7: Which model selection favours ( $\alpha$ -Rank,  $\beta = 1$ ). A node is a one-model population, larger when it gets more stationary mass; an arrow points to the model whose newcomer takes over more easily. Top:  $N = 6$ , so agents are compared with their tablemates. Bottom:  $N = 30$ , so mostly with agents at other tables. Under each graph: the self-play target rate, weighted by mass.**

grid, and each stays on top in at least 93% of bootstrap samples. Under weak selection ( $\beta = 0.1$ ) no model gets more than 0.34 of the weight within a table, so the cost of tablemate scoring needs selection that is at least moderately strong. The same agents, scored in two reasonable ways, lead to opposite outcomes for the group, in line with theory on selection between groups [27], and tablemate scoring is one simple route to the selfish outcomes that imitation produced for model-written strategies [31].

## 8 Discussion

No model grades its payments by the size of the risk under the base prompt. Four models pay a default of their own, close to the fair share or well above it, and still pay at  $p = 0$ , where paying helps no one, so generosity alone cannot explain their play. GPT Luna stops at  $p = 0$  but pays the same share at every positive risk. The default does not come from normative wording. It appears when the goal is left unstated, and stating the goal removes most payment at zero risk without making most models weigh the risk. All five adjust to partners only in simple ways, mostly by stopping once the outcome is settled.

*Possible causes.* Our data record choices, not reasoning, so we offer hypotheses. First, the equal split may be a focal point, a choice everyone expects everyone else to make; models rebuild it without the hint, and a 50/50 split can likewise reflect a focal point rather than a preference [6]. Second, a question may call up computation and a move request a default, which fits reports that models can hold a correct belief or plan and act against it [7, 23]. Third, post-training may favour doing one’s share when nothing says otherwise, which fits the finding that models are more cooperative than people [15] and our finding that naming the goal removes most payment at zero risk. Stronger reasoning is not an obvious fix, since reasoning models free-ride more in public goods games [10].

*Consequences for design. State the agent’s goal.* An agent that is told only the rules falls back on a default, and one sentence naming its goal removed the payment at zero risk that neutral wording left in place. Evaluations should report what the agent was told to want. *Test against the optimum, not the success rate.* A group can reach its target in every game and still waste most of its money, so a study needs  $\text{opt}(p)$ , a condition where paying is dominated, such as  $p = 0$ , and a check for payment after the outcome is settled. *Test in mixed populations.* Self-play hides how defaults interact: an exact fair-share default is optimal among copies of itself and fragile next to one late dropout, while a generous default absorbs that partner and pays for others. *Choose the selection signal with care.* Any leaderboard, market or imitation process that scores an agent against its tablemates favours whoever pays least, by Proposition 2, and in our games this lowered group success, while scoring across independent groups favoured the agent whose groups succeed. Each test needs only the game rules, a few scripted policies or a second model, and each caught a failure that self-play success rates did not show.

*Threats to validity.* The goal sentence is itself an instruction, and we test one phrasing of it, at three risk levels, in self-play only. Two other features of the setup may push models toward paying, and neither is tested here. The decision prompt asks for a final line with the move, so models may deliberate less when deciding than when answering a question. And the game is well known, so the fair share may be recalled rather than derived; the neutral wording removes the game’s name, but the rules still describe it. The same prompts were used for every model, so these features cannot explain why the models differ, but they may raise the common level of payment.

*Limitations.* We test five low-cost models, one per provider, in English and in one game; more capable models may respond to risk [3], so our claims are about this tier. Cells hold ten games, which limits power for the noisiest model, Grok. Our rewordings test wording, not every prompt design. The welfare losses assume risk-neutral players; the two tests where paying is dominated do not. The scripted partners, mixed tables and selection analysis use the base prompt only; mixed tables hold two models with fixed seat blocks within a cell, and  $\alpha$ -Rank assumes rare entry. For four models the self-play grid was collected separately; a rerun of the same prompt matches it (Table 3). Model text is not reproducible, so we release prompts, configurations, seeds and code, and promise reproducibility of the design rather than of the transcripts.

*Ethics.* The study involves no human subjects and no personal data. Its findings describe failure modes of deployed systems, and we report them so that designers can test for them before agents are trusted with shared resources.

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